Econ 527 HW4

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1 The Huggett Model

We solve the infinite-horizon Huggett (1993) model, in which households trade a single asset (bonds) in zero net supply. Each household maximises

$$\max_{\{a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to the period budget constraint and borrowing limit

$$c_t + a_{t+1} = (1+r)a_t + \exp(z_t), \quad c_t \geq 0, \quad a_{t+1} \geq a_{\min},$$

where idiosyncratic log-income follows the AR(1) process

$$\log z_t = \rho \log z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2),$$

and the period utility function is CRRA,

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}.$$

The Bellman equation is

$$V(a,z) = \max_{a' \geq a_{\min}} \; u((1+r)a + \exp(z) - a') + \beta\, \mathbb{E}\big[V(a', z') \mid z\big]. \tag{1}$$

Bonds are in zero net supply, so equilibrium requires

$$\mathcal{A}(r) \;:=\; \sum_i \sum_j a_i\, \phi(a_i, z_j) = 0,$$

and the equilibrium interest rate r^* satisfies $\mathcal{A}(r^*) = 0$.

Parameters are set as follows: $\gamma = 2$, $\beta = 0.96$, $\rho = 0.9$, $\sigma = 0.2$, $n_z = 7$, $n_a = 150$, grid curvature $\kappa = 2$. The borrowing limit is set to average income $a_{\min} = -\bar{z}$, where $\bar{z} = \mathbb{E}[\exp(z)] \approx 1.30$, and the upper bound is $a_{\max} = 50\bar{z}$. The AR(1) income process is discretised using Rouwenhorst's method.

1.1 Algorithm

We solve the household problem using Value Function Iteration (VFI) with the Endogenous Gridpoint Method. Starting from the initial guess $V^0(a_i, z_j) = u(Ra_i + \exp(z_j) - a_{\min})/(1-\beta)$, each outer iteration performs one full Bellman maximisation (exploiting monotonicity and concavity of V in a to speed the grid search), followed by $M = 30$ Howard improvement steps under the fixed policy. This reduces the total iteration count by roughly an order of magnitude relative to standard VFI.

The equilibrium is found by bisection on $r \in [-0.05,\ 1/\beta - 1 - \varepsilon]$. At each candidate rate, the household problem is solved, the stationary distribution ϕ is computed via the eigenvalue method (unit left eigenvector of the state-transition matrix Υ, normalised to sum to one), and net asset demand $\mathcal{A}(r)$ is evaluated. Bisection terminates when $|r_{\text{high}} - r_{\text{low}}| < 10^{-4}$.

1.2 Results

The equilibrium interest rate is $r^* \approx -0.044$. The negative rate reflects the precautionary savings motive: households accumulate assets even at negative real returns, so the bond market clears below the risk-free rate $1/\beta - 1 \approx 0.042$ that would prevail under complete markets. The gap between r^* and $1/\beta - 1$ measures the precautionary savings wedge.

../programs/figs/Value_Function_graph.png

Figure 1: Value function $V(a, z)$ for each income state z_j. The function is increasing and concave in a; higher income states shift it uniformly upward.

../programs/figs/Policy_fxn_graph.png

Figure 2: Savings policy function $g(a, z)$ for low and high income states, with the 45-degree line $a' = a$ shown dashed. Households with assets below the crossing point dissave; above it they accumulate. Higher income states shift the policy function upward.

../programs/figs/Phi_Dist_graph.png

Figure 3: Stationary distribution $\phi(a, z)$ for low and high income states. The distribution is concentrated near the borrowing limit with a thin right tail, typical of incomplete-market models.

The lower end of the wealth grid (the borrowing limit $a_{\min} \approx -1.30$) is binding for a non-trivial fraction of households: the precautionary motive is not strong enough to keep all agents away from the constraint at this interest rate. The upper end is not binding; the policy function lies strictly below $a_{\max}$ for all (a_i, z_j) and the stationary distribution has negligible mass near $a_{\max}$.

2 The Aiyagari Model

The Aiyagari (1994) model extends the Huggett framework to a production economy. Households face the same problem as above, but z_t now represents idiosyncratic log-labour supply and factor prices are determined by a representative firm. The budget constraint becomes

$$c_t + a_{t+1} = (1+r)a_t + w\exp(z_t), \quad c_t \geq 0, \quad a_{t+1} \geq 0,$$

with borrowing limit set to zero. A representative firm with production function $F(K, L) = K^{\alpha}L^{1-\alpha}$ rents capital and labour competitively, implying

$$r = \alpha K^{\alpha-1}L^{1-\alpha} - \delta, \qquad w = (1-\alpha)K^{\alpha}L^{-\alpha}.$$

Aggregate labour is $L = \mathbb{E}[\exp(z)]$ under the stationary distribution of the income process. Inverting the first firm FOC gives

$$K(r) = \left(\frac{\alpha L^{1-\alpha}}{r + \delta} \right)^{\frac{1}{1-\alpha}},$$

and equilibrium capital K^* satisfies

$$K^* = \mathcal{A}(r^*, w^*) \ := \ \sum_i \sum_j a_i \, \phi(a_i, z_j).$$

Additional parameters are $\alpha = 0.36$, $\delta = 0.10$, $a_{\max} = 60$; all other parameters are as in the Huggett model.

2.1 Algorithm

Equilibrium is found by bisecting on the capital stock K directly. At each candidate K, factor prices are computed from the firm FOCs, the household problem is solved (using the same VFI + Howard algorithm as above), and aggregate asset demand $\mathcal{A}(r(K), w(K))$ is evaluated. The excess demand function

$$\mathrm{ED}(K) = \mathcal{A}(r(K), w(K)) - K$$

is decreasing in K (higher capital lowers r, weakening the saving incentive), so bisection is well-defined. We terminate when $|K_{\text{high}} - K_{\text{low}}| < 10^{-4}$.

Since the household solver treats $\exp(z)$ as labour income internally, we rescale the income grid as $\tilde{z} = z + \log w$ before passing it to `f_spe`, so that $\exp(\tilde{z}) = w \exp(z)$ gives the correct wage income without modifying the solver.

2.2 Results

The table below summarises equilibrium outcomes. The equilibrium interest rate r^* lies below the risk-free rate $1/\beta - 1$ by the precautionary savings wedge; relative to the complete-markets steady state the Aiyagari economy features a higher capital stock because the precautionary motive drives additional saving.

Statistic	Symbol	Value
Equilibrium capital	K^*	*[from code]*
Equilibrium interest rate	r^*	*[from code]*
Equilibrium wage	w^*	*[from code]*
Output	Y^*	*[from code]*
Capital-output ratio	K^*/Y^*	*[from code]*
Market clearing error	$\mathcal{A} - K^*$	≈ 0

Table 1: Aiyagari equilibrium outcomes.

../programs/figs/Aiyagari_ValueFunction.png

Figure 4: Value function in the Aiyagari equilibrium for low, median, and high income states. As in the Huggett model, V is increasing and concave in a.

../programs/figs/Aiyagari_PolicyFunction.png

Figure 5: Savings policy function $g(a, z)$ in the Aiyagari equilibrium. High-income households save above the 45-degree line; low-income households at the natural borrowing limit $a_{\min} = 0$ save exactly zero.

../programs/figs/Aiyagari_WealthDist.png

Figure 6: Marginal stationary wealth distribution $\phi(a) = \sum_j \phi(a, z_j)$ in the Aiyagari equilibrium. The distribution is right-skewed with a mass point at $a_{\min} = 0$.

../programs/figs/Aiyagari_Lorenz.png

Figure 7: Lorenz curve for wealth in the Aiyagari equilibrium.

The Gini coefficient and top wealth shares are computed from the marginal distribution $\phi(a)$. The Gini is

$$\text{Gini} = 1 - 2\int_0^1 L(F)\,\mathrm{d}F,$$

where $L(F)$ is the Lorenz curve, evaluated numerically via the trapezoidal rule. Top-p% shares are $1 - L(1 - p/100)$.

Statistic	Model	US Data
Gini coefficient	*[from code]*	≈ 0.78
Top 10% share	*[from code]*	≈ 0.73
Top 1% share	*[from code]*	≈ 0.33

Table 2: Wealth inequality in the Aiyagari model versus US data (Wolff, 2017; Survey of Consumer Finances).

Standard Aiyagari models with CRRA utility and an AR(1) income process typically generate Gini coefficients of 0.35–0.45, well below the US value of ≈ 0.78. The discrepancy arises because the model abstracts from features important for the upper tail, including entrepreneurial risk (Quadrini, 2000), life-cycle heterogeneity (Huggett, 1996; De Nardi, 2004), luxury bequest motives (De Nardi, 2004), and heterogeneous returns to capital (Benhabib et al., 2019).

The upper bound $a_{\max} = 60$ is not binding: the policy function lies strictly below $a_{\max}$ and the stationary distribution has negligible mass there, confirming the grid is sufficiently wide. The lower bound $a_{\min} = 0$ is binding for low-income households, as expected.

3 Description of Code

Please run `PS4_hugget.m` for the Huggett model and `PS4_aiyagari.m` for the Aiyagari model. Both scripts cache intermediate results in a `results/` folder; delete this folder to rerun from scratch. Figures are saved to `figs/`.

3.1 Main Files

File	**Description**	**Dependencies**
`PS4_hugget.m`	Main script for Huggett model: sets parameters, runs bisection to find r^*, solves model at equilibrium, plots value function, policy function, and stationary distribution, and checks grid binding	`f_spe.m`, `f_nad.m`, `rouwenhorst.m`
`PS4_aiyagari.m`	Main script for Aiyagari model: computes aggregate labour L, bisects on K to find K^*, solves model at equilibrium prices, computes Gini coefficient and top wealth shares, and plots all figures	`f_spe.m`, `f_nad.m`, `rouwenhorst.m`

3.2 Solver Functions

File	Description	Dependencies
`f_spe.m`	`f_spe.m`	Stationary Partial Equilibrium solver. Takes income grid, transition matrix, utility handles, parameters, and asset grids; returns value function V, savings policy g, stationary distribution ϕ, and net asset demand $\mathcal{A}$. Implements VFI with $M = 30$ Howard improvement steps.
`mc_invdist` (local)		
`f_nad.m`	Net Asset Demand wrapper. Calls `f_spe` and returns only $\mathcal{A}(r) = \sum_{i,j} a_i \phi(a_i, z_j)$; useful for bisection routines that require a scalar-valued function	`f_spe.m`

3.3 General Functions

File	Description	Dependencies
`rouwenhorst.m`	Rouwenhorst's method for discretising an AR(1) process; returns the grid $z \in \mathbb{R}^{n_z}$ and transition matrix $\Pi \in \mathbb{R}^{n_z \times n_z}$	
`mc_invdist.m` (local in `f_spe`)	Computes the invariant distribution of a Markov chain as the normalised unit left eigenvector of Π; issues a warning if the unit eigenvalue is not unique	

4 Acknowledgments

Thanks to Professor Greaney for providing essential functions and for being very helpful throughout the course.

5 AI Usage

AI (specifically Claude) was used to debug code and format LaTeX.